Regulatory Acceleration and the Composition of FDA Approvals

Progress Report #2: Pivot to Controlled-Substance Approvals

Alejandro De La Torre

ECON 580

March 10, 2026

Pivot and New Research Questions

- **Old direction:** "Faster FDA approval $\rightarrow$ diversion risk" was too diffuse and hard to identify cleanly.
- **Primary question:** Did **PDUFA (1992)** increase approvals of **controlled substances**?
- **Secondary question:** Are **expedited review programs** disproportionately used for controlled substances?
- **Why this is stronger:** PDUFA is a policy shock, and controlled-substance status is an observable outcome.

Motivation and Mechanism

- **PDUFA**: user fees + review deadlines + expanded FDA capacity $\Rightarrow$ faster approvals.
- Existing work asks: **speed vs. safety** and launch timing.
- My pivot asks: did regulatory acceleration change **which types of drugs** enter the market?
- Controlled substances are a policy-relevant category with clear legal classification.

What the Literature Already Does

A) FDA speed / safety / PDUFA

- PDUFA and approval speed / safety tradeoffs (Philipson et al., 2008; Carpenter et al., 2008; Olson, 2008, 2009).
- Drugs reaching market sooner and launch timing (Dranove & Meltzer, 1994; Olson, 2009).

B) Institutional FDA history

- Descriptive regulatory history (Darrow et al., 2020).

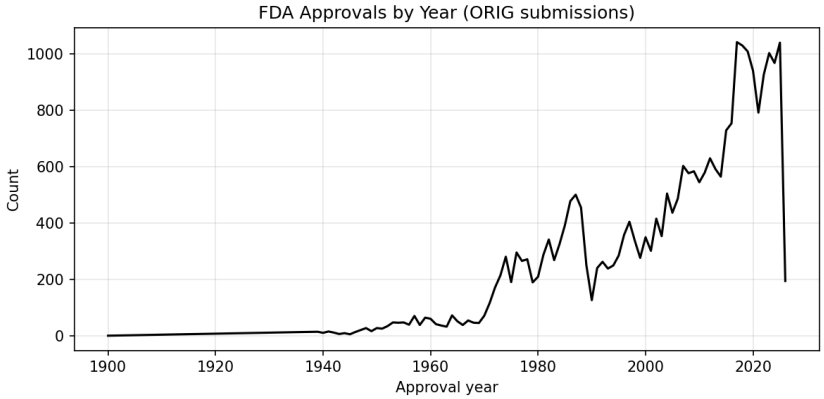
C) Supply-side drug policy / substitution

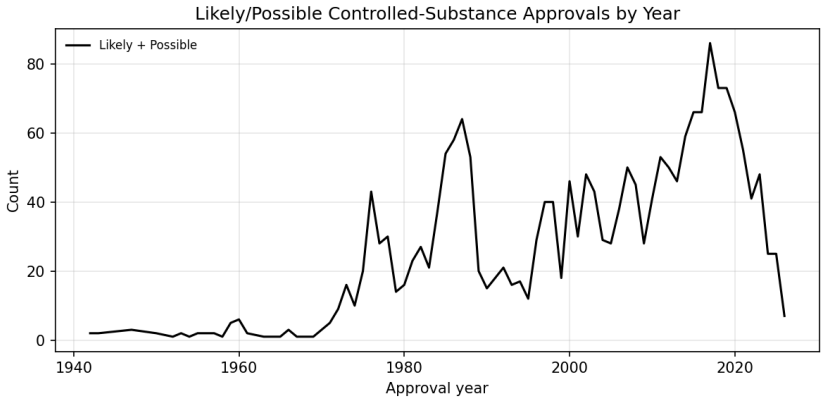
- Supply shocks and substitution in opioid markets (Alpert, Powell & Pacula, 2018; Powell & Pacula, 2021; Dobkin & Nicosia, 2009).

What Is New in This Project

- Existing work focuses on **speed vs. safety** and post-market outcomes.
- I focus on whether **regulatory acceleration changes composition** of approvals.
- Outcome is **controlled-substance status**, a clean, policy-relevant classification.
- Secondary contribution: whether expedited pathways are used disproportionately for drugs with abuse potential.

What I Have Explored So Far (Preliminary)





FDA approvals by year (Drugs@FDA).

Controlled-substance *candidates* by year.

Note: controlled-substance classification is currently a *heuristic screen* based on ingredient names; DEA schedule validation is pending.

Working Empirical Strategy (Draft)

- **Unit of observation:** drug-level FDA approvals over time (ORIG submissions).
- **Main outcome:** count/share/probability of controlled-substance approvals pre- vs post-PDUFA.
- **Secondary outcome:** expedited review usage within controlled substances.
- **Planned Figure 1:** controlled-substance approvals over time with PDUFA marked.
- **Planned Table 1:** summary by era and review designation.

Main Concerns / Open Identification Questions

- How should "controlled substance" be defined consistently over time (DEA schedules I–V vs subset)?
- Best comparison: levels, shares, or approval probabilities?
- How to handle changes in therapeutic mix over time?
- Should the framing be "PDUFA increased controlled-substance approvals" or "PDUFA changed composition"?
- What is the minimal defensible identification strategy for this class of question?

Next Steps + Feedback

Next steps (next 2–4 weeks):

1. Build validated FDA $\times$ DEA schedule match.
2. Refine descriptive figures and summarize by pre/post PDUFA eras.
3. Decide final outcome metric (counts vs shares vs probabilities).
4. Draft empirical design and outline data limitations.

Feedback request:

- Is the pivot persuasive and the contribution clear?
- Which controlled-substance definition is most credible for this question?
- Are there key papers or data pitfalls I should address early?

Comments and feedback welcome.